

Table 1: Summary of Existing Research on Automatic Passenger Counting and Crowd Monitoring Systems

Paper	Objective	Contributions	Results	Limitations	Model
A Field Study of Internet of Things-Based Solutions for Automatic Passenger Counting	Evaluate passenger counting technologies under real operating conditions.	Introduced a trial methodology for comparing APC solutions and provided practical insights for transport authorities.	Accuracy exceeded 70% in many cases, with best performance between 89–100%. Weekday and weekend performance varied.	Sensor installation issues, vibration sensitivity, reduced accuracy during mass boarding, and Wi-Fi blockage in dense crowds.	MobileNet SSD
A Low-Cost Automatic People-Counting System at Bus Stops Using Wi-Fi Probe Requests and Deep Learning	Estimate waiting passengers at bus stops using wireless devices.	Collected Wi-Fi probe requests and applied time-window packet analysis.	CRNN achieved the best counting performance.	Environmental noise, limited counting radius, and MAC randomization affect performance.	CRNN, LSTM
A Social Distance Estimation and Crowd Monitoring System for Surveillance Cameras	Estimate interpersonal distance and crowd density.	Developed a pose-estimation-based monitoring approach.	Localization strategy outperformed baseline methods.	OpenPose requires high computational resources.	OpenPose
Advances in Crowd Counting and Density Estimation Using Convolutional Neural Networks	Review CNN-based crowd counting techniques.	Presented CNN approach categories and summarized available datasets.	Reported overall accuracy of 95%.	Performance decreases under rapid lighting changes.	CNN
Label Noise Robust Crowd Counting with Loss Filtering Factor	Improve crowd counting with noisy labels.	Analyzed spatial and quantity noise effects.	$\theta = 0.9$ improved DMCC accuracy.	Requires hyperparameter tuning and higher computational cost.	VGG-19
Deep Convolutional Neural Network-Based Enhanced Crowd Density Monitoring for Intelligent Urban Planning on Smart Cities	Improve smart city crowd monitoring.	Used ConvLSTM for temporal patterns and DenseNet with SE blocks for feature extraction.	Achieved peak accuracy of 98.40%.	Struggles with extreme occlusion conditions.	ConvLSTM
A Wi-Fi-Based Automatic Bus Passenger Counting System	Track passenger numbers while preserving privacy.	Handled MAC randomization using anonymous passenger tracking.	Achieved 100% indoor accuracy and nearly 94% real-world accuracy.	Errors occur near close bus stops and inactive Wi-Fi devices.	De-randomization
NAPC: A Neural Algorithm for Automated Passenger Counting in Public Transport on a Privacy-Friendly Dataset	Provide real-time vehicle load information.	Used low-resolution 3D LiDAR data and introduced a dataset with 13,000 annotated sequences.	Obtained exact counts in about 96% of cases.	Requires extensive annotation effort and excludes children.	LSTM RNN
Video-Based Automatic People Counting for Public Transport: On-Bus Versus Off-Bus Deployment	Compare video APC systems for buses.	Evaluated low-cost video APC deployment and environmental effects.	Off-bus accuracy reached 88.9%; on-bus accuracy reached 75%.	Affected by weather, sunlight, occlusion, and accumulated errors.	MobileNetV3 SSD
Estimation of the Number of Passengers in a Bus Using Deep Learning	Estimate bus occupancy using deep learning.	Applied density estimation using a proprietary passenger image dataset.	Reduced MAE from 4.93 to 1.98 and performed well under different lighting conditions.	Corner-seat passengers may be missed.	YOLOv3, CAE